CMSC 414
Computer and Network Security

Lecture 14

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Mandatory access control

"Military security policy"

- Primarily concerned with secrecy
- Objects given "classification" (rank; compartments)
- Subjects given "clearance" (rank; compartments)
- "Need to know" basis
  - Subject with clearance (r, C) dominates object with classification (r', C') only if $r' \leq r$ and $C' \subseteq C$
  - Defines a partial order … classifications/clearance not necessarily hierarchical

Security models

- Multilevel security
  - Bell-LaPadula model
    - Identifies allowable communication flows
    - Concerned primarily with ensuring secrecy
  - Biba model
    - Concerned primarily with “trustworthiness”/integrity of data
- Multilateral security
  - Chinese wall
    - Developed for commercial applications

Bell-LaPadula model

- Simple security condition: S can read O if and only if $l_o \leq l_s$
- *-property: S can write O if and only if $l_s \leq l_o$
  - Why?
- "Read down; write up"
  - Information flows upward
- Why?
  - Trojan horse
  - Even with the right intentions, could be dangerous…

Basic security theorem

- If a system begins in a secure state, and always preserves the simple security condition and the *-property, then the system will always remain in a secure state
  - I.e., information never flows down…

Communicating down…

- How to communicate from a higher security level to a lower one?
  - Max. security level vs. current security level
  - Maximum security level must always dominate the current security level
  - Reduce security level to write down…
    - Security theorem no longer holds
    - Must rely on users to be security-conscious

Commercial vs. military systems

- The Bell-LaPadula model does not work well for commercial systems
  - Users given access to data as needed
    - Discretionary access control vs. mandatory access control
  - Would require large number of categories and classifications
  - Centralized handling of "security clearances"

Biba model

- Concerned with integrity
  - "Dual" of Bell-LaPadula model
- The higher the level, the more confidence
  - More confidence that a program will act correctly
  - More confidence that a subject will act appropriately
  - More confidence that data is trustworthy
- Integrity levels may be independent of security classifications
  - Confidentiality vs. trustworthiness
  - Information flow vs. information modification

Biba model

- Simple integrity condition: S can read O if and only if $I_s \leq I_o$
  - I_s, I_o denote the integrity levels
- (Integrity) *-property: S can write O if and only if $I_o \leq I_s$
  - Why?
  - The information obtained from a subject cannot be more trustworthy than the subject itself
- "Read up; write down"
  - Information flows downward

Security theorem

- An information transfer path is a sequence of objects $o_1, \ldots, o_n$ and subjects $s_1, \ldots, s_{n-1}$, such that, for all i, s_i can read o_i and write to o_{i+1}
  - Information can be transferred from o_1 to o_n via a sequence of read-write operations
- Theorem: If there is an information transfer path from o_1 to o_n, then $I(o_n) \leq I(o_1)$
  - Informally: information transfer does not increase the trustworthiness of the data
- Note: says nothing about secrecy…

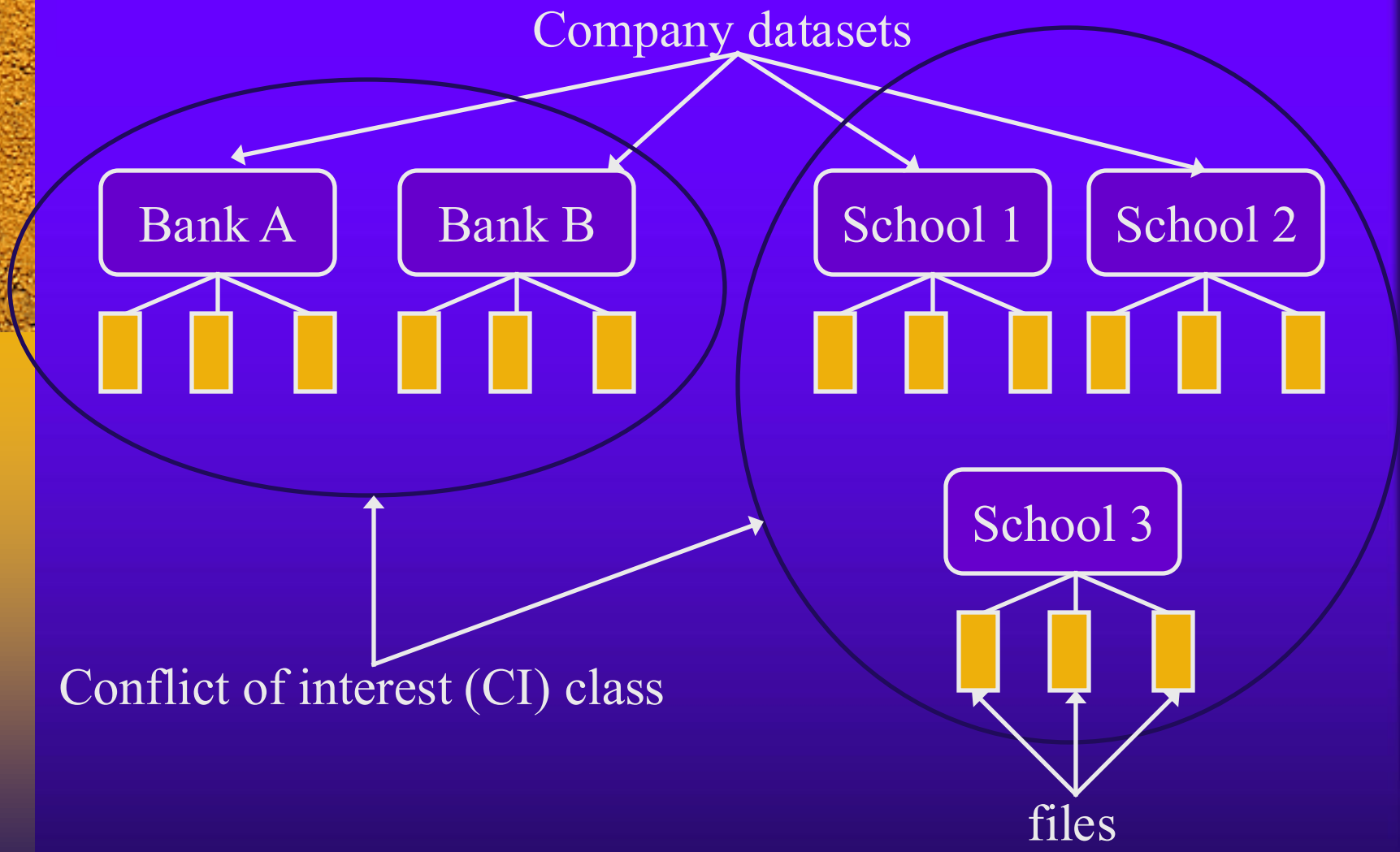
"Low-water-mark" policy

- Variation of "pure" Biba model
- If s reads o, then the integrity level of s is changed to $\min(l_o, l_s)$
  - The subject may be relying on data less trustworthy than itself
- If s writes to o, the integrity level of o is changed to $\min(l_o, l_s)$
  - The subject may have written untrustworthy data to o
- Drawback: the integrity level of subjects/objects is non-increasing!

Chinese wall

- Intended to prevent conflicts of interest
- Rights are dynamically updated based on actions of the subjects

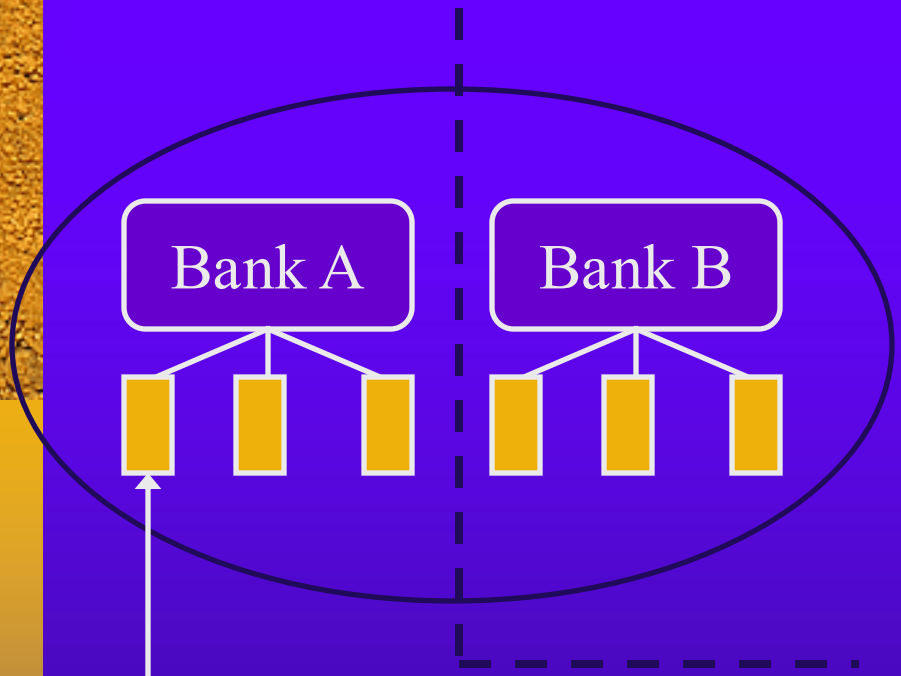
Chinese wall -- basic setup

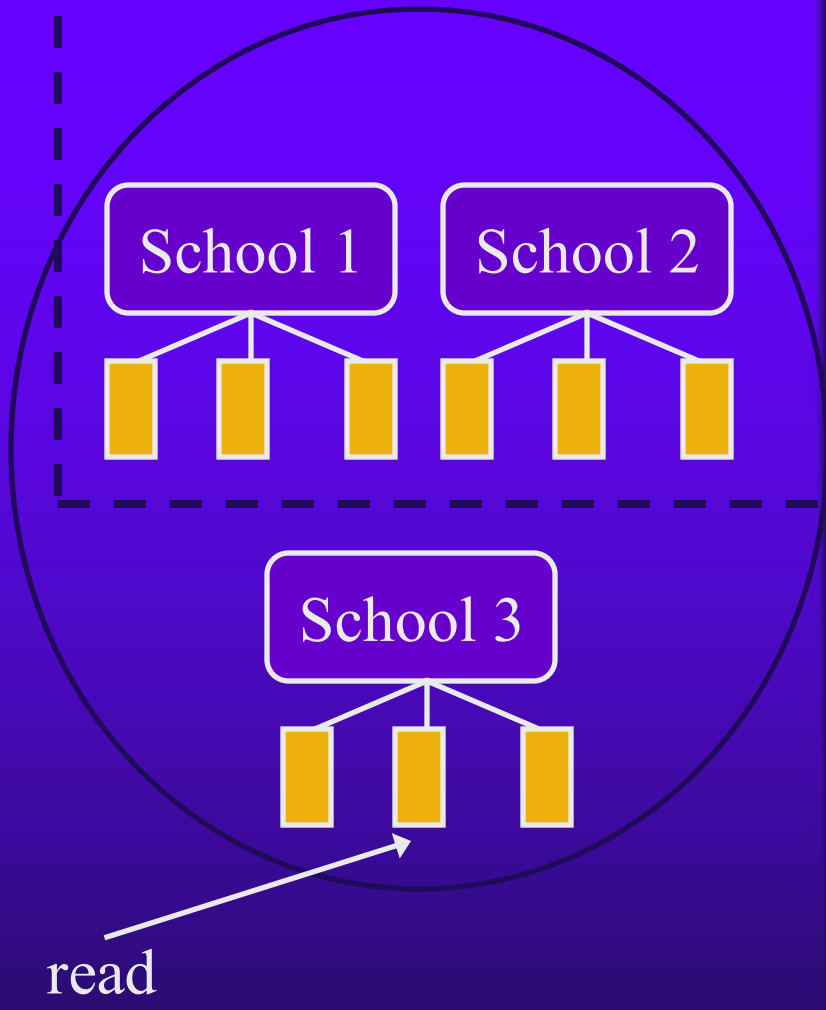


Chinese wall rules

- Subject S is allowed to read from *at most one* company dataset in any CI class
  - This rule is dynamically updated as accesses occur
  - See next slide…

Example





Chinese wall rules II

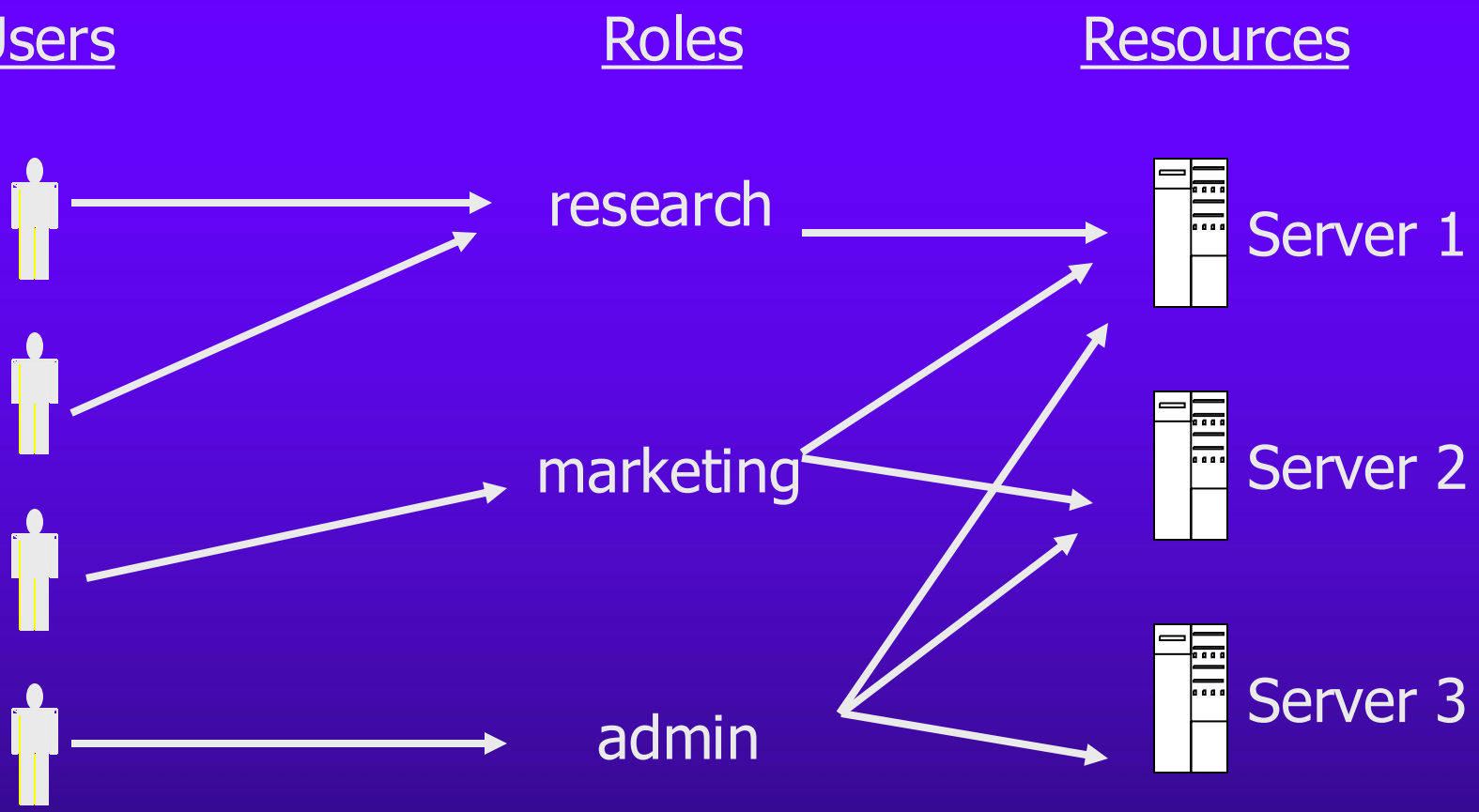
- S can write to O only if
  - S can read O and
  - All objects that S can read are in the same dataset as O
- This is intended to prevent an indirect flow of information that would cause a conflict of interest
  - E.g., S reads from Bank A and writes to School 1; S' can read from School 1 and Bank B
  - S' may find out information about Banks A and B!
- Note that S can write to at most one dataset…

Role-based access control

RBAC

- Access controls assigned based on *roles*
  - Can use an access matrix, where "subjects" are roles
- Users assigned to different roles
  - Can be static or dynamic
  - A user can have multiple roles assigned
  - Can use "access matrix" with users as rows, and roles as columns
  - Will, in general, be more compact than a full-blown access control matrix
- Advantage: users change more frequently than roles

RBAC: basic idea



Questions…

- Where might each of DAC, MAC, or RBAC make the most sense?

Code-based access control

Identity-based vs. code-based

- The access control policies we have discussed so far have all been *identity-based*
  - I.e., ultimately decisions come down to the identity of the principal/subject
- This works in ‘closed’ organizations
  - Principals correspond to known people
  - Organization has authority over its members
  - Users can be held accountable for their actions
- Does not work in ‘open’ settings
  - E.g., running code from the web

Code-based access control

- Determine rights of a process based on characteristics of the code itself, and/or its source
  - E.g., code downloaded from local site or remote site?
  - E.g., code signed by trusted source?
  - E.g., does code try to read from/write to disk?
  - E.g., does code contain buffer overflows?
    - Checked locally
    - 'Proof-carrying code'

Difficulties

- Difficulties arise when one process calls another
  - E.g., remote process calls local process, or signed process calls an unsigned process
- Case 1: “trusted” g calls “untrusted” f
  - Default should be to disallow access
  - But g could explicitly delegate its right to f
- Case 2: “untrusted” f calls “trusted” g
  - Default should be to disallow access
  - But g could explicitly ‘assert’ its right
  - (cf. confused deputy problem)

Java 1 security model

- Unsigned applets limited to sandbox
  - E.g., no access to user's filesystem
- Local code unrestricted
  - Since Java 1.1, signed code also unrestricted
- Drawbacks
  - No finer-grained control
  - Code location not follproof
    - Local filesystem on remote machine
    - Remote code that gets cached on the local machine

Java 2 security model

- *Byte code verifier, class loaders*
- *Security policy*
  - Grants access to code based on code properties determined by the above
- *Security manager/access controller*
  - Enforce the policy

Byte code verifier

- Analyzes Java class files (using, e.g., static type checking and data-flow analysis) to ensure certain properties are met
- E.g.,
  - No stack overflow
  - Methods called with arguments of appropriate type
  - No violation of access restrictions
- Note: these are static checks, not run-time checks

Class loaders

- Link-time checks performed when needed classes are loaded

Security policy

- Maps attributes of the code to permissions
  - Developers may define application-specific permissions
- May depend on the source code itself, as well as any code signers

Security manager

- The ‘reference monitor’ in Java
- Invoked at run-time to check the *execution context* (i.e., execution stack) against required permissions
  - Each method on the stack has a class; each class belongs to a *protection domain* indicating permissions granted to the class
- Security manager computes the intersection of permissions for all methods on the stack (‘stack walk’), and compares against required permissions
  - A method can also *assert* permissions, in which case prior callers are ignored

An example

```
f(){
  foo;
  g(); }

g(){
  bar;
  doPrivileged…
  h(); }

h(){
  … }
```

read, /tmp

h	read, /tmp
g	read, /tmp
f	

$\mathbf{Perms = Perm_h \cap Perm_g \cap Perm_f}$

Trusted Computing

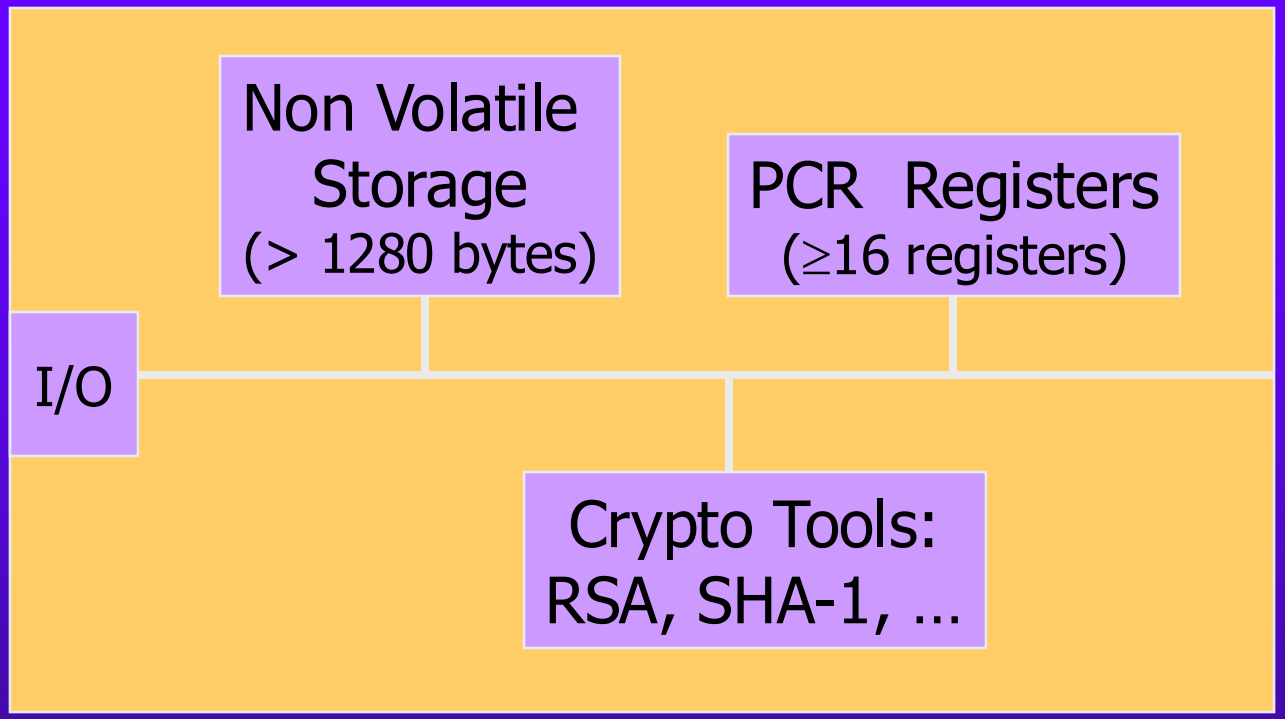
Overview

- Secure hardware (*Trusted Platform Module*, or TPM) installed in computer
- Goals
  - Secure boot
  - Software verification
  - Attestation
  - Encrypted storage
- This is already deployed

Disclaimer

- The intent of the following is to give the high-level ideas, rather than completely correct low-level details
- Full specification available on-line
  - TCG consortium

TPM chip


Non Volatile Storage (> 1280 bytes)
PCR Registers (≥16 registers)
I/O
Crypto Tools: RSA, SHA-1, …

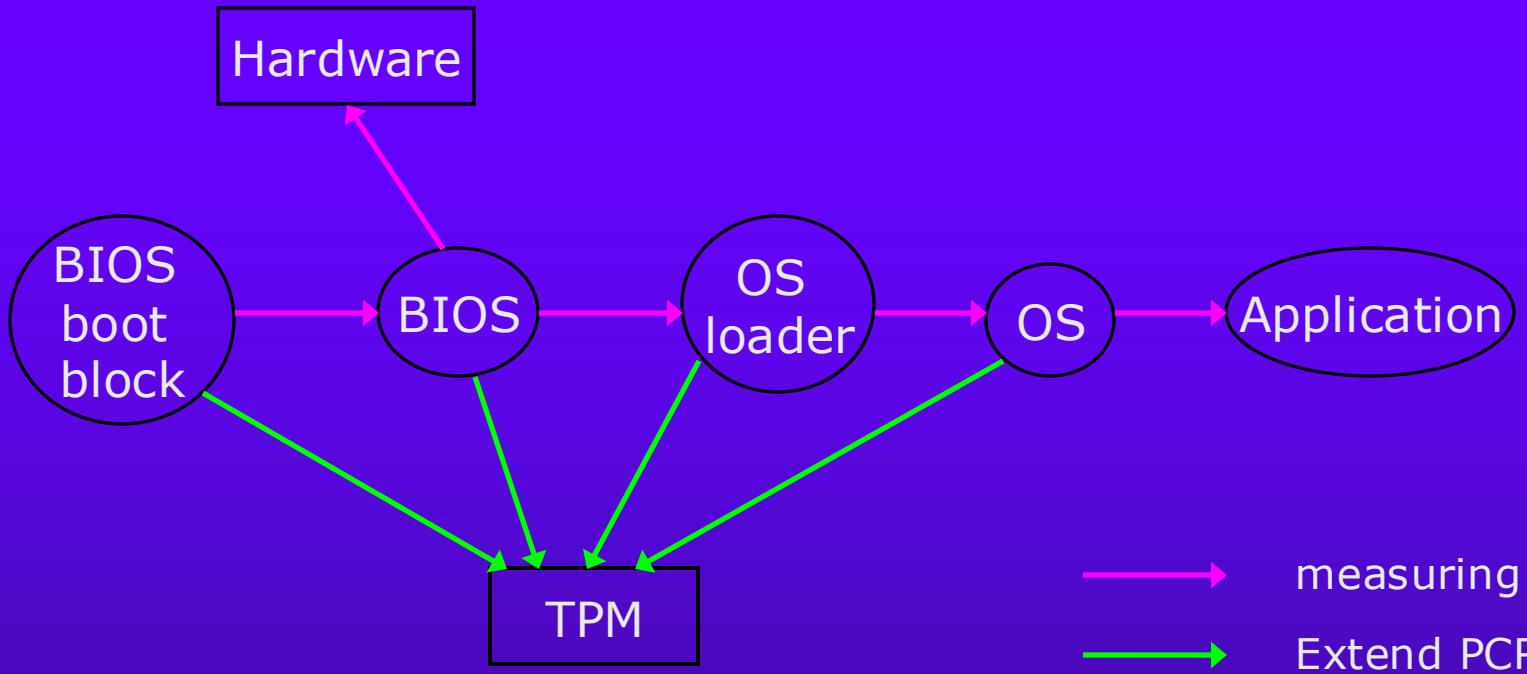
Non-volatile storage

- Endorsement keys (EK) [RSA]
  - Created at manufacturing time, bound to computer
  - Signing keys; used for attestation
  - Cannot be changed (enforced by hardware)
  - Tamper-resistant; user cannot read or modify EK
- Storage root key (SRK) [RSA]
  - Created by user; can be changed
  - Used to encrypt data

PCR

- "Platform Configuration Registers"
- 20 bytes; hold SHA-1 output
- Can only be modified in two ways (enforced by the hardware):
  - TPM_Startup (initialize the contents of the PCR)
  - TPM_Extend(D): PCR = SHA-1 (PCR || D)
- Used to obtain an "image" of the loaded software…

PCM usage



- Collision resistance of SHA1 ensures "uniqueness"

What is this good for?

- Compare computed value with reference value
  - Secure boot
- Software validation
  - Verify signature before installing new software
  - All this verifies is the source
- Decrypt data
  - Decrypt only if in known (good) configuration
- Attestation
  - Prove to a third party that you are in a good configuration

Encrypted data

- Encrypt AES key K with SRK; encrypt bulk data with K
  - Hybrid encryption!
- When encrypting the AES key, embed current PCR value
  - E.g., $\text{Sign}_{EK}(\text{PCR}, \text{Enc}_{SRK}(K))$
  - (This is not actually the way it is done)
- When decrypting, check that the embedded value matches the current value
  - Refuse to decrypt if this is not the case!
- Can also incorporate a user password, etc.

Attestation

- Goal: prove to a remote party what software is running on my machine
- Applications:
  - Prove to company network that no viruses are running on my machine
  - Prove to another player that I am running an unmodified version of Quake
  - Prove to Apple that I am running iTunes…

Basic idea

- Sign PCR value with EK
  - Actually, sign with attestation identity key (AIK) validated with EK (ignore this for now)
- Assume third party knows EK
  - There is actually a PKI
- To prevent replay, use nonce provided by the third party
- Third party verifies signature; verifies that PCR corresponds to known “good” state

Controversy

- Loss of anonymity
  - Signature using EK uniquely identifies the machine it came from
    - Some recent crypto proposals to address this
  - Third parties can tell what software you are running
- Loss of control
  - What if google says you need to have google desktop installed in order to use their search engine?
  - What if Sony says you must use their music player to download their music?
  - User can't access information on the machine they own